

### Wandi-SSAL Bound Value Capability Design

The Wandi-SSAL design uses bound values for many of its capabilities. For example, by establishing a bound value of 64 characters length for log message string argument, the design checks and verifies that string arguments do not exceed 64 characters. This mitigates problems where string pointer is not properly NULL terminated.

These and other bound values ensure that the logger API functions are robust, reliable and prevent abnormal behavior impacting the normal execution flow on the application.

Detailed examples on how the Bound Argument Validation values works with the LoggerAppsMsg() API are provided in the Wandi-SSAL-examples.c program.

Some other Dynamic Adjustable Parameter capability examples include:

| Dynamically Adjustable Parameters                                               |                                                                                                                                              |                                                                                                        |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Bound Definition                                                                | Bound Values                                                                                                                                 | Benefits                                                                                               |
| Number of format string specifiers types per log message                        | Maximum number of format string specifiers types allowed per message is 6.                                                                   | Verify that log message arguments do not exceed an unreasonable number of arguments.                   |
| Different types of format string specifiers.                                    | Different types of supported format string specifiers are %s, %d, %f, %c, %x, %p.                                                            | Prevent use of format spring specifiers not in the list and block malicious attacker specifier format. |
| Length for string arguments.                                                    | Maximum length for string %s argument is 64 characters.                                                                                      | Avoid abnormal behavior due to unbounded or invalid string length.                                     |
| Length for message format string specification.                                 | Maximum length format string specification is 256 characters.                                                                                | Avoid abnormal behavior due to unbounded or invalid string length.                                     |
| Subset of ASCII characters for format string specification and string argument. | Fixed subset of ASCII characters allowed:<br>upper/lower case alphanumeric, blank , \\, %, \, \" and !#\$&'()*+,-./:;<=>?@[^_`{ } characters | Avoid characters not needed for message logging, such as non-printable characters.                     |

|                                                                               |                                                                                             |                                                            |
|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Number of severity id=value pairs and length of value string.                 | Maximum number of severity levels is 4 and length of string value is 16 characters.         | Use only valid pre-defined severity level data.            |
| Number of logical message grouping id=value pairs and length of value string. | Maximum number of logical message groups is 32 and length of string value is 16 characters. | Use only valid pre-defined logical message groups.         |
| Multiple sequential log files each with N messages per file.                  | Define N to be the maximum number of lines per log file.                                    | Generate multiple log files instead of one very long file. |